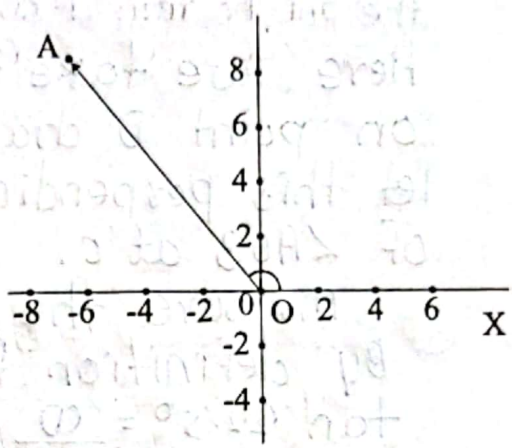
(1) Construct an angle in standard position whose terminal arm passes through A(-6, 8). Hence find all trigonometric ratios of this angle.

**Solution :** $OA = r = \sqrt{(-6)^2 + (8)^2} = \sqrt{100} = 10$




$A \equiv (-6, 8) \equiv (x, y)$

$\therefore\ x = -6$ and $y = 8$

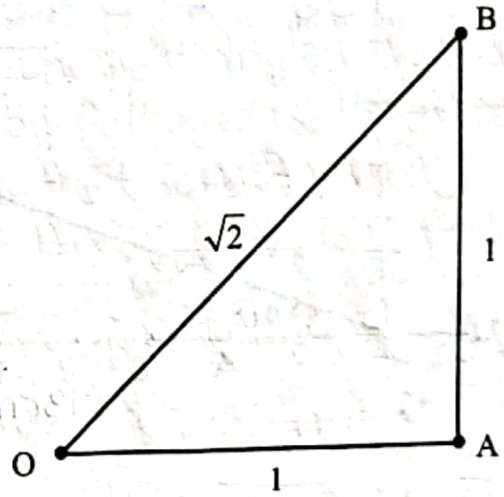
Let measure of $\angle XOA$ be $\theta$.

$\sin\theta = \frac{y}{r} = \frac{8}{10} = \frac{4}{5}$ $\qquad$ $\operatorname{cosec}\theta = \frac{1}{\sin\theta} = \frac{5}{4}$

$\cos\theta = \frac{x}{r} = \frac{-6}{10} = -\frac{3}{5}$ $\qquad$ $\sec\theta = \frac{1}{\cos\theta} = -\frac{5}{3}$

$\tan\theta = \frac{y}{x} = \frac{8}{-6} = -\frac{4}{3}$ $\qquad$ $\cot\theta = \frac{1}{\tan\theta} = -\frac{3}{4}$

(2) Construct angle of measure 22.5° by bisecting angle of measure 45°. Hence find the value of tan 22.5°




**Solution :** Let Δ OAB be an isosceles right angle triangle and OA= AB = 1.

$\therefore$ OB=$\sqrt{2}$

Draw bisector of angle AOB.

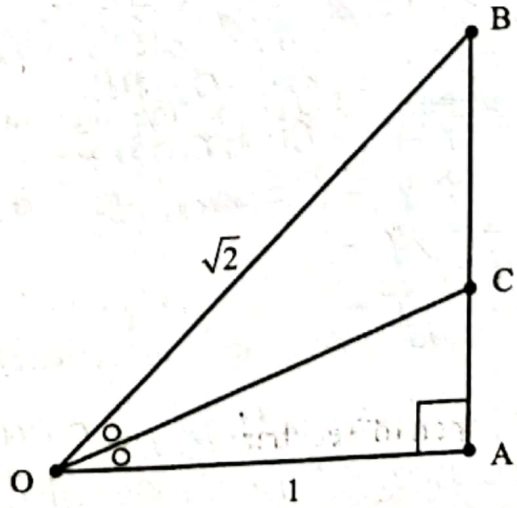
Let it intersect AB in C.

$m\angle AOC = 22.5^0$

Note that C is not the mid point of AB.

Use section formula to find length of AC.

Use Pythagoras theorem to find OC. Then write trigonometric ratios of angle $22.5^0$. Write your calculations here.



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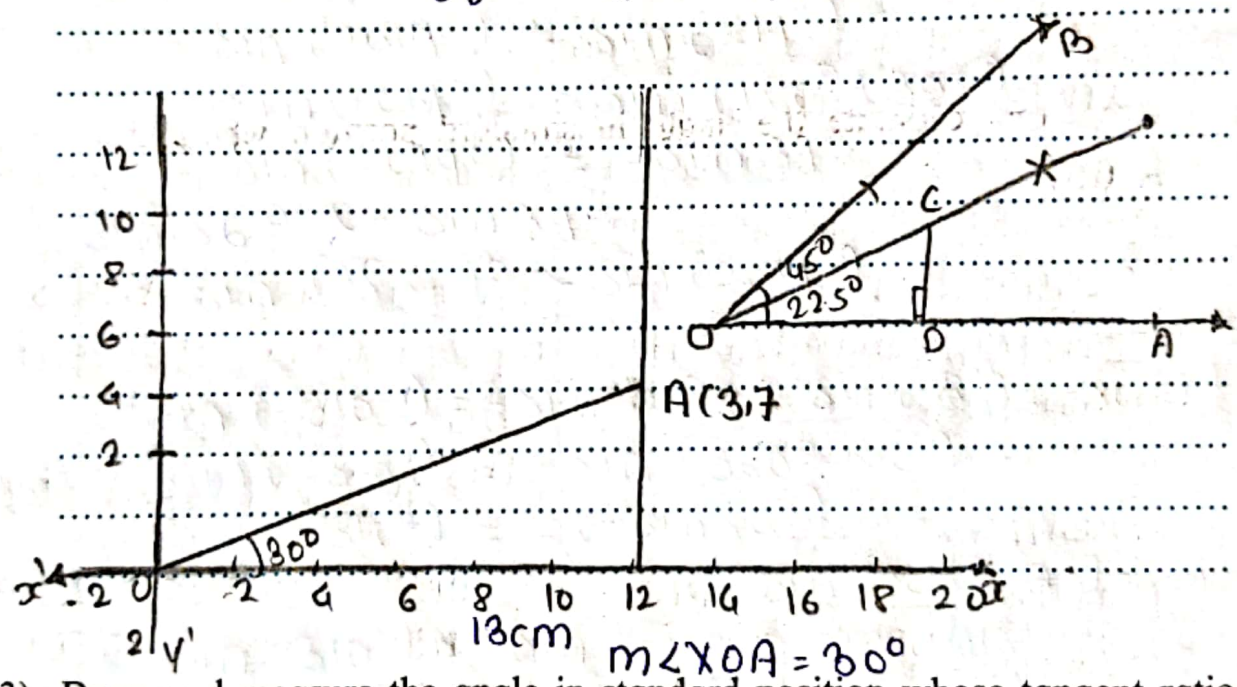
Draw $\angle AOB = 45^\circ$ using protractor.
Bisect $\angle AOB$ on ray OA we can take OD of any length.

Here, we take OD = 4 cm
on point D draw perpendicular to OA
let this perpendicular meets the bisector of $\angle AOB$ at C.
measure the length CD here CD = 1.7 cm
By definition from $\Delta OCD$
$\tan 22.5^\circ = \frac{CD}{OD} = \frac{1.7}{4} = \frac{17}{40} = 0.0425$




(3) Draw and measure the angle in standard position whose tangent ratio is $\frac{7}{13}$.

**Solution :** Let θ be the measure of one of the angles having tangent ratio $\frac{7}{13}$

As $\tan\theta = \frac{y}{x} = \frac{7}{13}$, the terminal arm of the angle passes through A(13, 7)
Now draw angle in standard position whose terminal arm passes through A(13,7). Using protractor measure the angle XOA.



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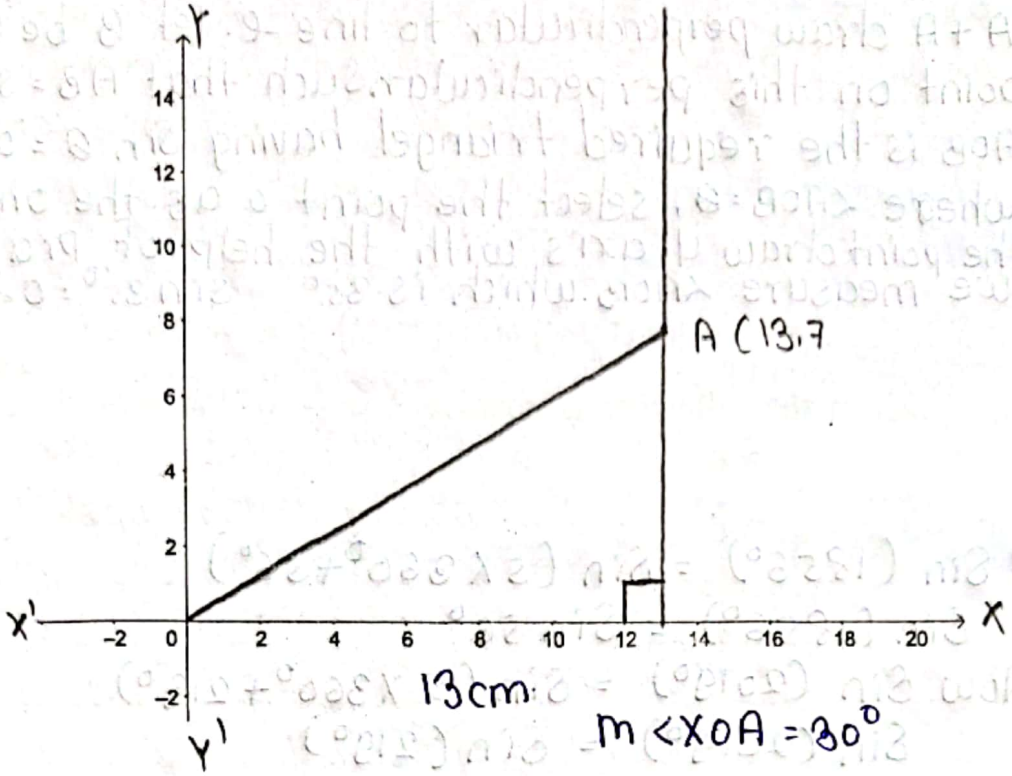
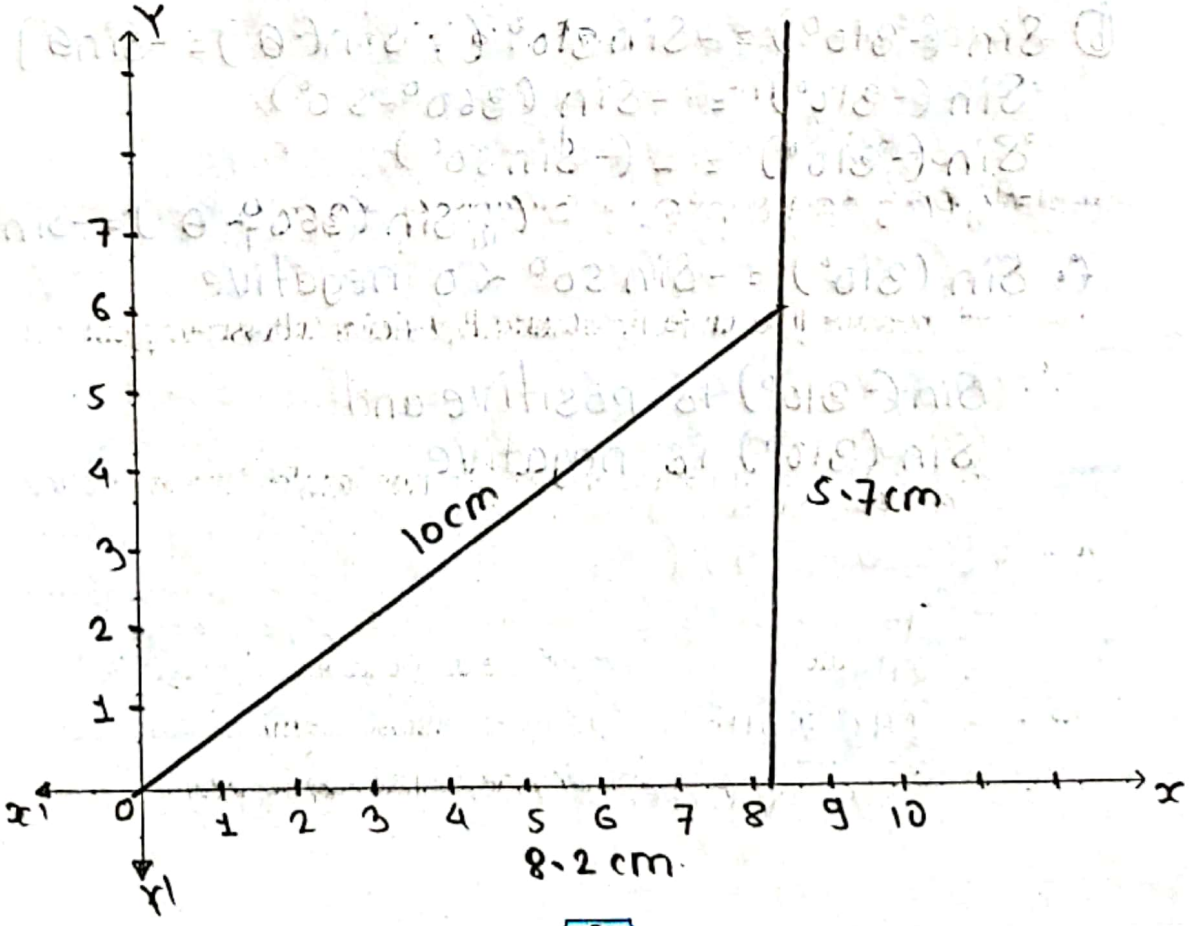



(4) Draw and estimate the angle in standard position whose sine ratio is 0.57.

**Solution :** Draw a line l and take any point A on line l



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A + A draw perpendicular to line e. let B be any point on this perpendicular such that AB = 5.7 cm, AOB is the required triangel having $\sin\theta = 0.57 = \frac{5.7}{10}$ where $\angle AOB = \theta$, select the point O as the origin. Through the point draw y-axis with the help of protractor. we measure $\angle AOB$ which is $35^\circ$ $\sin 35^\circ = 0.57$

(5) (a) Which is greater $\sin(1856^\circ)$ or $\sin(2019^\circ)$ ? Give reason.

(b) Which of the following is positive ? $\sin(-310^\circ)$ or $\sin(310^\circ)$ ? Give reason.

**Solution :**

(a) $\sin(1856^\circ) = \sin(5 \times 360^\circ + 56^\circ)$

$\sin(1856^\circ) = \sin 56^\circ$

Now $\sin(2019^\circ) = \sin(5 \times 360^\circ + 219^\circ)$

$\sin(2019^\circ) = \sin(219^\circ)$

$\sin(2019^\circ) = \sin(180^\circ + 39^\circ)$

$\sin(2019^\circ) = -\sin 39^\circ$

$\sin 56^\circ > -\sin 39^\circ$

$\therefore \sin(1856)^\circ > \sin(2019^\circ)$

(b) $\sin(-310^\circ) = -\sin 310^\circ$ $[\because \sin(-\theta) = -\sin\theta]$

$\sin(-310^\circ) = -\sin(360^\circ - 50^\circ)$

$\sin(-310^\circ) = -(-\sin 50^\circ)$

$[\because \sin(360^\circ - \theta) = -\sin\theta]$

$\because \sin(310^\circ) = -\sin 50^\circ < 0$ negative

$\therefore \sin(-310^\circ)$ is positive and

$\sin(310^\circ)$ is negative



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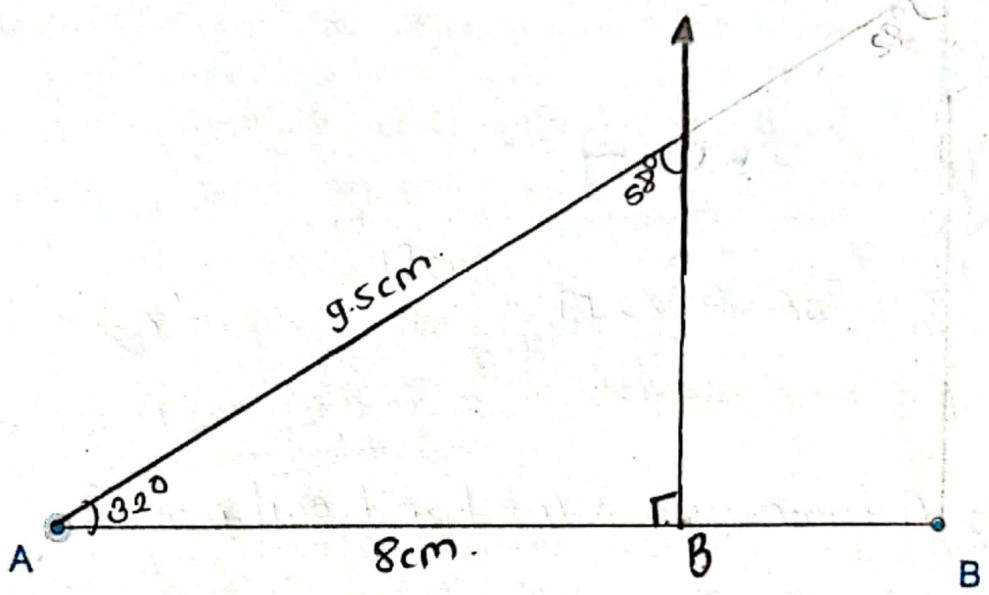
(6) Construct a triangle, the measures of whose angles are $32^0, 58^0$ and $90^0$. Measure the lengths of sides of this triangle. Find $\sin 32^0$ and $\cos 32^0$. Hence find the value of $\sin^2 32° + \cos^2 32°$.

**Solution :** Construct $\Delta$ ABC with A = 32°, B = 90°, C = 58°.




we Construct a triangle having measures of angle $32^0, 58^0, 90^0$

we take base AB of $\Delta ABC$

AB = 8 cm

At point B draw $\perp$ AB

using pratractor draw an angle of measure $32^0$ at A

let the side of this angle cats the perpendicular at C.

This is the required $\Delta ABC$ from the triangle.

$\sin 32^0 = \frac{5.1}{9.5}$, $\cos 32^0 = \frac{8}{9.5}$

$\sin^2 32^0 + \cos^2 32^0$

$1 = \left(\frac{5.1}{9.5}\right)^2 + \left(\frac{8^2}{9.5}\right)^2$

$= \left[\frac{5.1}{(19/2)}\right]^2 + \left[\frac{8}{(19/2)}\right]^2$

$= \frac{26.01 \times 4}{361} + \frac{64 \times 4}{361}$

$= \frac{104.4 + 256}{361} = \frac{360.04}{361} = 0.997$

$= 1$ (approx)

Sign of Teacher :



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